ETH zürich

Chair of Cognitive Science

Representing Design Intention and User Experience:
A Node-Based Cognitive Walkthrough & 360° Behavioural Experiment Pipeline

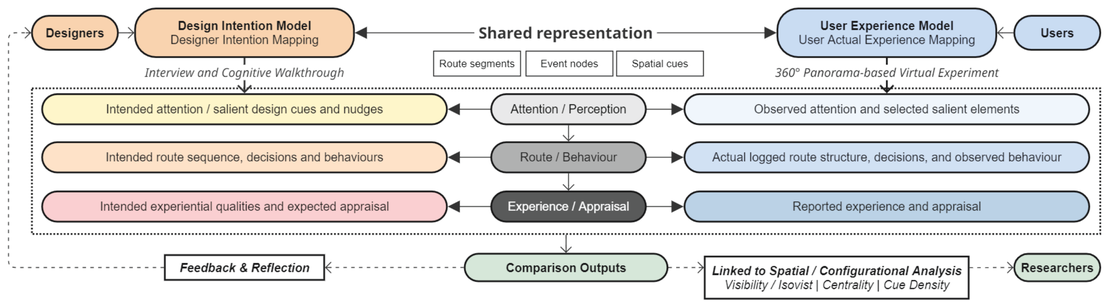
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1 Introduction

Architects compose spaces and circulation logic throgh spatial nudges such as salient architectural features, views, and route structure to suggest how people should look, move and feel. However, mismatches may occur between designers' intended logic and users' actual experience, which lead to usability problems. There is also a representational gap where designers reason from global bird's eye views, while users encounter buildings sequentially from local viewpoints, shaped by situated affordances and cues.

We propose a comparative framework and node-based pipeline for aligning intended and actual user journeys through shared spatial units across behaviour, attention and experience. This allows discrepancies to be spatially located, analysed, and translated into evidence-based feedback for architects.

2 Comparative Framework for the Two Models



A comparative framework between a Design Intention Model (DIM) and a User Experience Model (UEM) with shared spatial units and representations across movement behaviour, attention and experience at both route and node levels.

2 Method Overview

A.Semi-structured Architect Interview



- Floorplans & models
- Design goals & concepts
- Main circulation logic
- Target users & scenarios
- Define task journeys
- Key areas of uncertainties

B.Environment Preperation

- Construct node-edge graph



- Capture panorama photo/rendering source
- Assign 360 to nodes



- Set directional chevrons in the scene

C.Experiment Set-up

Task mode
[navigation, exploration]

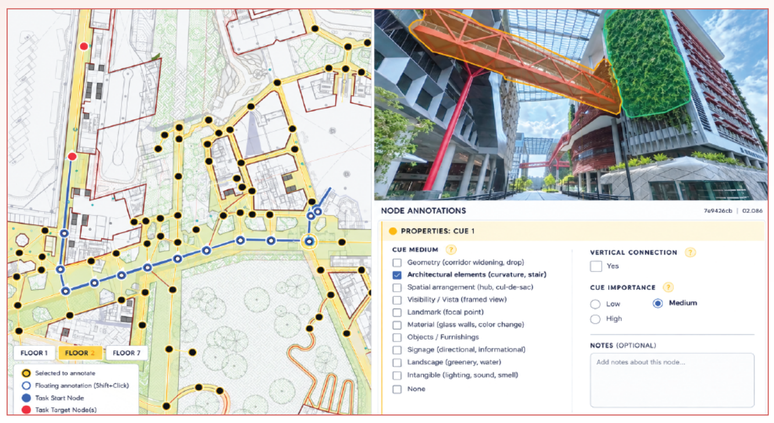
Task origin and destinations

Task instructions and limits

Design questionnaire

D.Design Intention Cognitive Walkthrough

Design intentions are captured and translated into a **structured route graph** with annotated event nodes, design cues, expected behaviour and experiential targets, through a **cognitive walkthrough** session. It also serves as a reflective exercise, making designers' circulation assumptions explicit.



Upload floorplan and node graph

Annotate design cues on the panorama [medium, importance, confidence]

Draw intended route path and key decision points

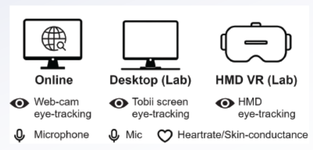
Elicit intended behaviour, attention, affect dimension and rationale behind

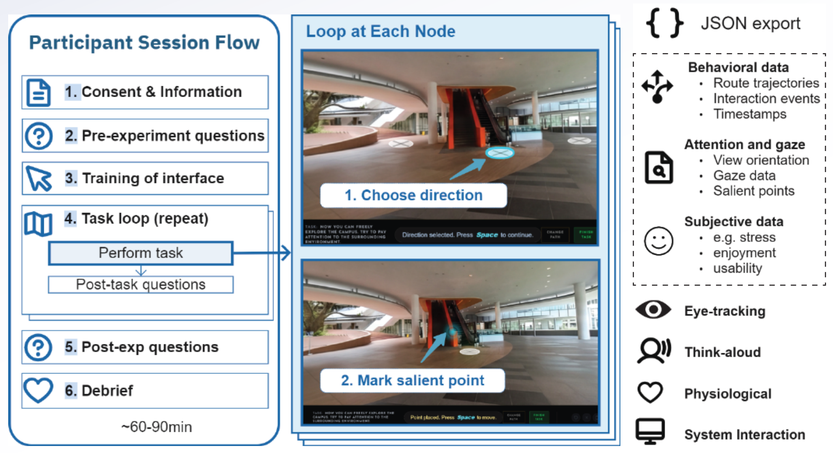
E.Users' Virtual Wayfinding Experiment

A **customisable behavioural experiment system** for researchers and architects, using node-based virtual 360° panoramas. It provides a practical middle ground between resource-intensive VR/field studies and lighter qualitative approaches, while capturing route choices, node visits, view behaviour, salient-point selections, task performance, and experiential ratings.

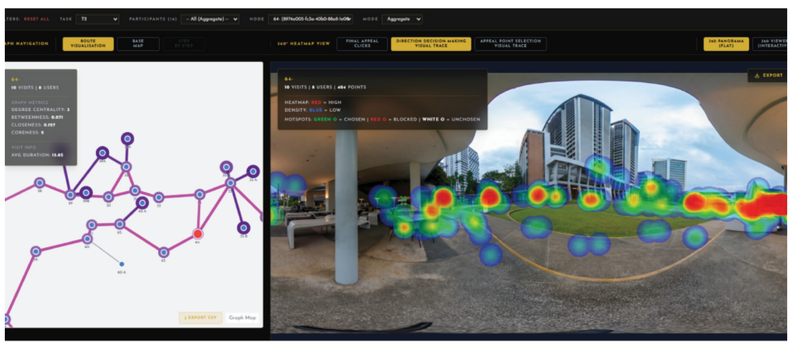
SCAN TO TRY!

Participant Mode Setup





3 Pilot Results

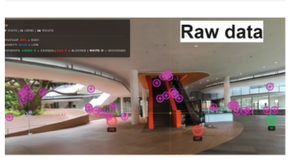


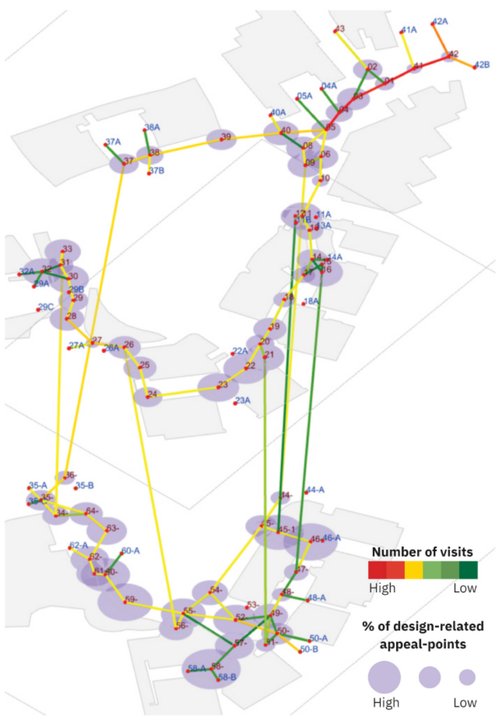
The pilot experiment at UTown, NUS, Singapore demonstrates how the pipeline translates 360° navigation and exploration task results into spatially located evidence. Route choices, node visits, salient-point selections, head-rotation, and experience ratings can be compared with the intended route and linked to configurational measures.

< Platform output and visualisation for route trajectories, node visits and view behavoiur.



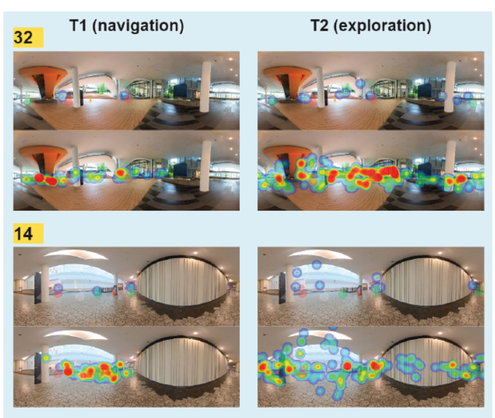


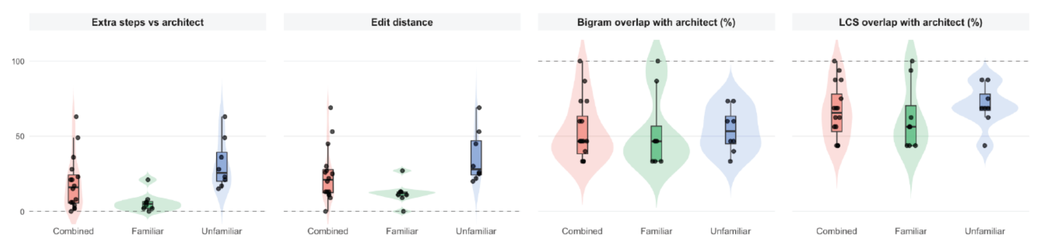




< Intended vs participant route raw data: The intended or ideal route is compared with aggregated participant routes and raw salient-point selections exported from the experiment platform.

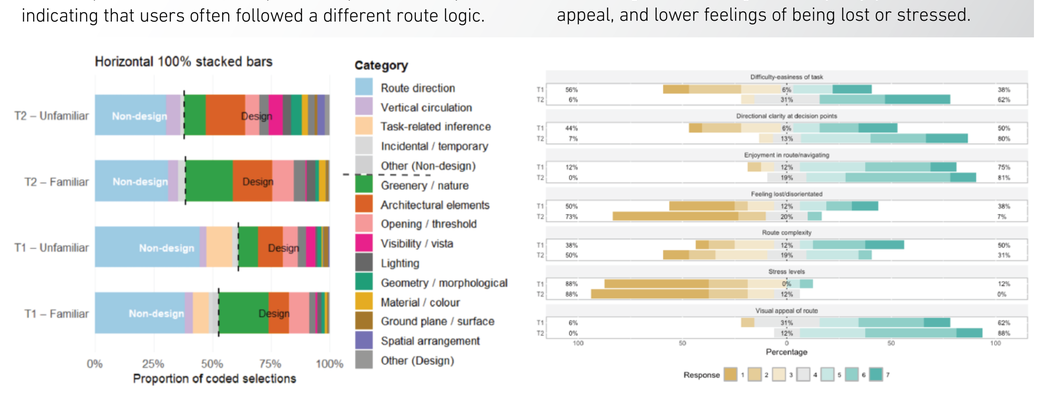
> Attention to salient features: Salient-point selections differed by task: navigation focused more on route continuation and task-relevant cues, while exploration produced broader attention to design-related features.





^ Route alignment / discrepancy: Participant routes showed clear divergence from the intended route, with extra steps, edit distance, and partial sequence overlaps indicating that users often followed a different route logic.

v Affect and experiential ratings: Post-task ratings suggest that exploration was experienced more positively than navigation, with higher clarity, enjoyment, and visual appeal, and lower feelings of being lost or stressed.



4 Contributions and Next Steps

1. A node-based comparative pipeline: Translates both intended and experienced journeys into shared spatial units: routes, nodes, edges, decision points, and cues.
2. A bridge between design intent and spatial analysis: Connects designers' intended circulation logic with observed movement, attention, and appraisal data.
3. An adaptable toolkit for human-centred design evaluation: Supports pre-occupancy testing, post-occupancy evaluation, cognitive walkthroughs, behavioural experiments set-up, and reflective feedback for architects.

- Scale up the workflow; Work with authoring design team; Integrate multi-modal data and translatedinto design feedback

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